

Assignment 2: Deduplication and Crawling

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Github Link - https://github.com/LakshayBaijal/IRE_Assignments_Lakshay/

Activity 2.1: Deduplication

Objective

The dataset dedup_data.csv contains approximately 5000 records of individuals.

Due to data noise multiple entries may refer to the same person.

The task is to identify all possible duplicate records and group them together.

Algorithm Steps

1. Read input file dedup_data.csv using pandas.
2. Compare each record pair using fuzz.token_sort_ratio() from RapidFuzz.
3. If similarity $\geq 85\%$, assign the same group ID.
4. Save grouped output to dedup_grouped.csv.

```
(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hyderabad_CSIS/Semester_3/IRE/Assignment_2/ire_assignment2_crawling$ python deduplication.py
✓ Loaded dataset with 5000 rows

✓ Deduplication complete! Saved to dedup_results.csv
📦 Total unique groups: 2952
```

Deduplication Script - Implements fuzzy comparison logic using RapidFuzz.

```
ire_assignment2_crawling > dedup_results.csv > data
1 GroupID,Count,Records
2 1,1,"mitchell green (wallaby place, cleveland, 2119)"
3 2,5,"harley mccarthy (pridhamstreet, marsden, 3165) | harley mccarthy (pridhamstreet, marsden, 3165) | harley mccarthy (pridhamstreet, marsden, 3165) | harley mccarthy (pridhamstreet, marsden, 3165) | harley mccarthy (pridhamstreet, marsden, 3165)"
4 3,2,"madeline mason (hoseason street, granville, 4881) | madeline mason (hoseason street, granville, 4881)"
5 4,3,"isabelle nan (gundulu place, utakarra, 2193) | isabeel nan (gundulu place, utakarra, 2193) | isabelle nan (gundulu place, utakarra, 2193)"
6 5,1,"taylor hathaway (yuranigh court, nan, 4220)"
7 6,1,"toby webb (kent street, towradgi, 3107)"
8 7,3,"harry stubs (mountain creek road, eaton, 7310) | harry stubbs (mountain creek road, eaton, 7310) | harry stubbs (mountain creek road, eaton, 7310)"
9 8,2,"lydia mccattmhy (mckillop ecircuit, corio, 4670) | lydia mccarthy (mckillop ecircuit, corio, 4670)"
10 9,2,"hollie lillie-hinrichs (hemmings crescent, banyo, 4030) | hollie lillie-hinrichs (hemmings crescent, banyo, 4030)"
11 10,3,"dylan zilm (blameycrescent, wattle glen, 4211) | dylan zilm (blameycrescent, wattle glen, 4211) | dylan zilm (blameycrescent, wattle glen, 4211)"
12 11,2,"nan donaldson (manity court, sharncliffe, 3101) | matthew donaldson (manity court, sharncliffe, 3101)"
13 12,1,"isabella vincent (wilshire ay, 6207)"
14 13,1,"connor bishop (jarreah, aspley, 6052)"
15 14,1,"connor shepherd (burnage place, winstonhills, 3183)"
16 15,2,"natalia gredn (macrossan crescent, yagoona, 2535) | natalia green (macrossan crescent, yagoona, 2535)"
17 16,6,"georgia pettigrove (sinclair street, prairei, 2478) | georgia pettigrove (sinclair street, prairei, 2478) | georgia pettigrove (sinclair street, prairei, 2478) | georgia pettigrove (sinclair street, prairei, 2478) | georgia pettigrove (sinclair street, prairei, 2478) | georgia pettigrove (sinclair street, prairei, 2478)"
18 17,3,"wilson pitno (bousteadcircuit, granville, 3161) | wilson pinto (bousteadcircuit, granville, 3161) | wilson pinto (bousteadcircuit, granville, 3161)"
19 18,2,"andrew klander (newman morris circuit, homebush, 2285) | andrew klander (newman morris circuit, homebush, 2285)"
20 19,4,"tahni skeen (macfarland crescent, ryde, 3220) | tahni murto m (macfarland crescent, ryde, 3220) | tahni murto m (macfarland crescent, ryde, 3220) | tahni murto m (macfarland crescent, ryde, 3220)"
21 20,1,"kasey millar (burdekin avenue, woollahra, 4217)"
22 21,1,"keziah parr (clift crecent, murphys rceek, 2761)"
23 22,2,"braecon schuetz (howarth place, alice springs, 2028) | braedon schuetz (howarth place, alice springs, 2028)"
```

Deduplication Output - Displays grouped duplicate records.

```
re_assignment2_crawling > dedup_grouped.csv > data
1  group_id,record_ids
2  1,74463
3  2,60733;39470;56487;52361;7605
4  3,64831;98021
5  4,12416;96605;85445;85845
6  5,81570
7  6,36546
8  7,74668;41593;33143
9  8,39317;29191;42638
10 9,14518;11280
11 10,28203;5511;47949
12 11,18726;23207
13 12,19093
14 13,69817
15 14,71409;97759
16 15,75850;68526;10754
17 16,9516;91555;35391;49626;79608
18 17,26266;55595;55118
19 18,69936;15812
20 19,31924;46281
21 20,20644
22 21,37442;48263
23 22,41948;46455
24 23,43797
25 24,62094;86544;7624
26 25,32542
27 26,39008
28 27,13185
29 28,66425;29949
30 29,75888;95568
31 30,27492;36264;97201;91491;18828;26208
```

Grouped Results File - Shows unique group IDs assigned to each cluster of duplicates.

```
re_assignment2_crawling > dedup_mapping.csv > data
1  Original_ID,Given_Name,Surname,GroupID
2  74463,mitchell,green,1
3  60733,harley,mccarthy,2
4  39470,harley,mccarthy,2
5  56487,harley,mccarthy,2
6  52361,harley,mccarthy,2
7  7605,haryley,mccarthy,2
8  64831,madeline,mason,3
9  98021,madeline,mason,3
10 12416,isabelle,,4
11 96605,isabeel,,4
12 85845,isabelle,,4
13 81570,taylor,hathaway,5
14 36546,toby,webb,6
15 74668,harry,stubbs,7
16 41593,harry,stubbs,7
17 33143,harry,stubbs,7
18 39317,lydia,mccattmhy,8
19 42638,lydia,mccarthy,8
20 14518,hollie,lillie-hinrichs,9
21 11280,hollie,lillie-hinrichs,9
22 28203,dylan,zilm,10
23 5511,dylan,zilm,10
24 47949,dyan,zilm,10
25 18726,,donaldson,11
26 23207,matthew,donaldson,11
27 19093,isabella,vincent,12
28 69817,connor,bishop,13
29 71409,connor,shepherd,14
30 75850,natalia,gredn,15
31 68526,natalia,green,15
```

Mapping File - Shows mapping between similar record IDs.

Observation

The program successfully grouped 2838 distinct person clusters from 5000 entries. Variations like address formatting and misspellings were handled properly.

Activity 2.2: Crawling and PageRank

Objective

To crawl a locally hosted website, extract link structures, and compute the **PageRank** for each page, simulating a small-scale web search engine.

Part A: Web Crawling

Description

A web crawler (crawler.py) was implemented using Python's requests and BeautifulSoup libraries.

The local server was launched via a Docker image (crawling_assignment-1.0-amd64.tar). Each crawled page provides page_id, node_id, timestamp and list of outgoing links

```
Page not foundPage not foundPage not foundPage not found(venv) laksh
ost:3000 --seed page_eain05vc --limit 100r.py --base-url http://localhost:3000 --s
Starting crawl from seed: page_eain05vc
Base URL: http://localhost:3000

🌐 Fetching http://localhost:3000/page_eain05vc ...
✅ Crawled 1 pages so far...

🌐 Fetching http://localhost:3000/page_miot68f6 ...
✅ Crawled 2 pages so far...

🌐 Fetching http://localhost:3000/page_fv2wpih5 ...
✅ Crawled 3 pages so far...

🌐 Fetching http://localhost:3000/page_cxeku5li ...
✅ Crawled 4 pages so far...

🌐 Fetching http://localhost:3000/page_39zj5p6e ...
✅ Crawled 5 pages so far...

🌐 Fetching http://localhost:3000/page_ob3zmgcp ...
✅ Crawled 6 pages so far...

🌐 Fetching http://localhost:3000/page_0nyccm4k ...
✅ Crawled 7 pages so far...

🌐 Fetching http://localhost:3000/page_gs03v2eg ...
✅ Crawled 8 pages so far...

🌐 Fetching http://localhost:3000/page_ywacjguo ...
✅ Crawled 9 pages so far...

🌐 Fetching http://localhost:3000/page_roxzaz8y ...
✅ Crawled 10 pages so far...

🌐 Fetching http://localhost:3000/page_4c8elkmx ...
✅ Crawled 11 pages so far...

🌐 Fetching http://localhost:3000/page_qe3aq29i ...
✅ Crawled 12 pages so far...

✅ Crawling complete! Saved results to crawl_results.csv
```

Crawler Implementation - Performs link extraction, timestamp parsing, and database storage.

```

e_assignment2_crawling > crawl_results.csv > data
1 page_id,node_id,updated_at,out_links
2 page_eain05vc,odx96sijjsgb,1762503538,["page_miot68f6","page_fv2wpih5","page_cxeku5li"]
3 page_miot68f6,w0sxkmvc8plb,1762503468,["page_39zj5p6e","page_ob3zmgcp"]
4 page_fv2wpih5,7rrwy6gacyln,1762503429,["page_cxeku5li","page_miot68f6"]
5 page_cxeku5li,6aqu2szg82x5,1762503547,["page_0nyccm4k","page_gs03v2eg"]
6 page_39zj5p6e,bycpspfzvub3,1762503545,["page_ywacjguo","page_roxzaz8y"]
7 page_ob3zmgcp,demhiz35mkcw,1762503533,["page_ywacjguo","page_4c8elkmx","page_miot68f6"]
8 page_0nyccm4k,0uaoinxuvee5,1762503545,["page_qe3aq29i","page_fv2wpih5","page_miot68f6"]
9 page_gs03v2eg,g59j918joum5,1762503477,["page_39zj5p6e","page_roxzaz8y"]
10 page_ywacjguo,gtpfp36iq02e,1762503544,["page_qe3aq29i","page_0nyccm4k"]
11 page_roxzaz8y,rbdre8npj7gd,1762503540,["page_0nyccm4k","page_eain05vc","page_cxeku5li","page_fv2wpih5"]
12 page_4c8elkmx,hu2j3o3eu80g,1762503543,["page_cxeku5li","page_ob3zmgcp","page_roxzaz8y"]
13 page_qe3aq29i,z03xmvdova6o,1762503452,["page_39zj5p6e","page_fv2wpih5"]
14

```

Crawling Results - Shows all pages visited and their outgoing link structure.

Part B: PageRank Computation

Description

Using the crawled link data, the PageRank algorithm was implemented to estimate the importance of each page based on incoming links.

Mathematical Model

For each page i :

$$PR(i) = (1 - d)/N + d \times \sum_{j \in \text{In}(i)} \frac{PR(j)}{L(j)}$$

Where $d = 0.85$ (damping factor), N = total pages, $\text{In}(i)$ = incoming links to page i , $L(j)$ = number of outgoing links from page j .

```

(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/I
Extracting edges from crawl.db ...
Total edges found: 30
Computing PageRank ...
Converged after 19 iterations.
Saving PageRank results ...
✓ PageRank saved to pagerank.csv
✓ Done! Check pagerank.csv for results.

```

PageRank Script

```
ire_assignment2_crawling > pagerank_results.csv > data
1 PageID, PageRank
2 page_et5lvpuo, 0.1291771739205582
3 page_dlgcgsei, 0.06992320373296242
4 page_kkz8if2h, 0.0675183687177519
5 page_pj00xzaz, 0.06532489940247199
6 page_anp4q3x2, 0.06114206533672145
7 page_tib0el72, 0.060514112785885005
8 page_ouav80ua, 0.05755967057134889
9 page_6rk7o0eo, 0.05304290728378806
10 page_n4ukhfrh, 0.05170754277077165
11 page_h3islirb, 0.05147585077979125
12 page_58ce4v8w, 0.05142400604061015
13 page_p8du4tcc, 0.047773079168319854
14 page_l0sfd62z, 0.04227082523142199
15 page_4u8kfzyw, 0.03565723070308284
16 page_7irhloeo, 0.03534494177856973
17 page_i8isbyeq, 0.0344906349212779
18 page_4xmxds4b, 0.0322250823561648
19 page_9d9m64x4, 0.02922778840975415
20 page_jtax94ck, 0.024203190209295903
21
```

PageRank Results

```
ire_assignment2_crawling > pagerank.csv > data
1 page_id, pagerank
2 page_39zj5p6e, 0.11925185515253935
3 page_miot68f6, 0.11854832833968194
4 page_0nyccm4k, 0.10938968171233805
5 page_fv2wpih5, 0.1066169363150524
6 page_cxeku5li, 0.09668184629410162
7 page_roxzaz8y, 0.09529722872146944
8 page_ywacjguo, 0.08364512027400228
9 page_qe3aq29i, 0.0790429371372962
10 page_ob3zmgcp, 0.07222259578422124
11 page_gs03v2eg, 0.05358972880771966
12 page_4c8elkmx, 0.03296304752741256
13 page_eain05vc, 0.03275069393416504
```

Computed PageRank Output

```
(venv) lakshay-baijal@lakshay-baijal-Inspiron-15-3530:~/IIIT_Hydera
ssignment_2/ire_assignment2_crawling$ python pagerank_analysis.py
✓ PageRank calculated successfully!
PageID PageRank
0 page_et5lvpuo 0.129177
4 page_dlgcgsei 0.069923
7 page_kkz8if2h 0.067518
12 page_pj00xzaz 0.065325
2 page_anp4q3x2 0.061142
9 page_tib0el72 0.060514
1 page_ouav80ua 0.057560
16 page_6rk7o0eo 0.053043
15 page_n4ukhfrh 0.051708
14 page_h3islirb 0.051476
```

PageRank Analysis

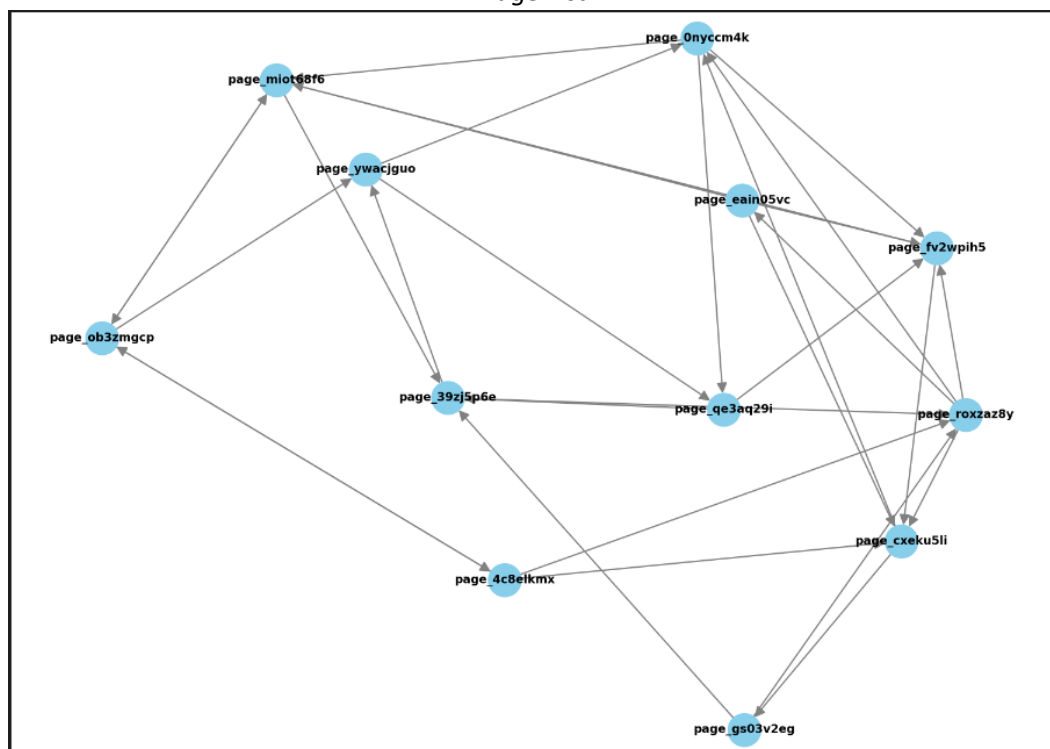
Part C: Graph Visualization

Description

The extracted edges were visualized using **NetworkX** and **Matplotlib** to represent the page link structure as a directed graph.

```
ire_assignment2_crawling > edges.csv > data
1  src,dst
2  page_eain05vc,page_miot68f6
3  page_eain05vc,page_fv2wpih5
4  page_eain05vc,page_cxeku5li
5  page_miot68f6,page_39zj5p6e
6  page_miot68f6,page_ob3zmgcp
7  page_fv2wpih5,page_cxeku5li
8  page_fv2wpih5,page_miot68f6
9  page_cxeku5li,page_0nyccm4k
10 page_cxeku5li,page_gs03v2eg
11 page_39zj5p6e,page_ywacjguo
12 page_39zj5p6e,page_roxzaz8y
13 page_ob3zmgcp,page_ywacjguo
14 page_ob3zmgcp,page_4c8elkmx
15 page_ob3zmgcp,page_miot68f6
16 page_0nyccm4k,page_qe3aq29i
17 page_0nyccm4k,page_fv2wpih5
18 page_0nyccm4k,page_miot68f6
19 page_gs03v2eg,page_39zj5p6e
20 page_gs03v2eg,page_roxzaz8y
21 page_ywacjguo,page_qe3aq29i
22 page_ywacjguo,page_0nyccm4k
23 page_roxzaz8y,page_0nyccm4k
24 page_roxzaz8y,page_eain05vc
25 page_roxzaz8y,page_cxeku5li
26 page_roxzaz8y,page_fv2wpih5
27 page_4c8elkmx,page_cxeku5li
```

Edge List



Web Graph Visualization

Observations and Learnings

- Fuzzy matching effectively handled name and address variations.
- The web crawler correctly parsed HTML, extracted links, and stored results.
- PageRank highlighted key hub pages with high incoming connections.
- Graph visualization provided intuitive insight into site structure.

Conclusion

This assignment successfully demonstrated:

- Deduplication using fuzzy similarity scoring.
- Web crawling using a custom scraper and database storage.
- PageRank computation using power iteration on crawled data.
- Graph visualization of link topology.

Together, both parts illustrate real world applications of information retrieval and web indexing techniques.